

Emotion-aware recommendation over WikiArt $\cap$ ArtEmis

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AI usage disclaimer

AI tools were used in two ways: as support for the implementation of the code, and as an aid in composing this report. Because the project was carried out individually, AI additionally served as a discussion partner for brainstorming and for working through the interpretation of certain results. Design decisions, and conclusions are mainly by the author.

1. What this project is and user scenario

The project presents two recommendation surfaces for an online art catalogue, built on one frozen corpus of 20,833 paintings that carries WikiArt’s artist and style labels together with ArtEmis’s crowd-annotated emotional responses. The application context is an online art catalogues deployed in the GLAM (Galleries, Libraries, Archives, Museums) sector, private art galleries, and commercial art trading. **Track A** is a personalized feed within a logged-in visitor scenario, answering “what should this user see next”; **Track B** is an item-to-item “More like this” module for an anonymous browser, answering “what else is like this painting”. Feedback is implicit and binary throughout: an item is shown, and it is either engaged with or it is not.

The two tracks share the corpus and nothing else — different queries, different relevance definitions, different candidate pools. **Their numbers are never comparable to each other.**

	Track A — “For you”	Track B — “More like this”
is there a user?	yes — simulated	no; only a painting on screen
feedback	implicit binary engagement — simulated	none; content and labels only
relevance	a held-out engaged item	another work by the same artist
candidates	catalogue minus the user’s seen set	catalogue minus the query (20,832)
queries	12,000 users	2,000 sampled paintings

Declaration — the user layer is synthetic. Neither WikiArt nor ArtEmis contains a user identifier, and collecting real preference data was out of scope. Everything on Track A runs on a simulated impression-and-engagement log, so every accuracy figure there describes *recovery of a known ground truth*, not performance on real visitors. Track B uses no synthetic data at all.

2. Data and feedback

WikiArt (81,444 images, Kaggle mirror steubk/wikiart) supplies images plus categorical artist and style labels, and a perceptual hash used for deduplication. Its `genre` column is unusable — it repeats the style label rather than holding a subject taxonomy — so it is dropped.

ArtEmis (~455K rows) supplies emotion attributions with free-text explanations from crowdworkers, **5.77 annotators per painting** on average in the final corpus, each picking one of nine classes (amusement, awe, contentment, excitement, anger, disgust, fear, sadness, *something else*). It joins to WikiArt on the image path; mojibake and apostrophe mismatches break ~2% of that join and are normalised, which also merges 5 artists split across two spellings.

Three signal types are kept strictly apart:

signal	what it is	real or simulated
content	style, artist, 9-class emotion distribution, pixels	real
elicited annotation	the emotion a paid annotator attributed to a painting	real , but <i>not</i> feedback
implicit feedback	impressions and binary engagement per (user, item)	simulated

ArtEmis is elicited annotation, not user feedback. Annotators were assigned artworks and paid to report a response; they did not choose what to view. It is therefore used as a content *feature* on both tracks, never as the feedback signal.

The simulated feedback layer is 12,000 users and ~2.1M impressions over the corpus (~100 per item), each with a binary outcome, plus a uniformly-exposed 20-item holdout per user. Two properties matter more than its size, and both are deliberate: **missing is not negative** — an item never shown carries no information, while an item shown and ignored is a weak negative — and **the log is missing-not-at-random**, because exposure is proportional to item popularity *and* to the user’s own latent affinity. The log therefore over-represents both famous paintings and paintings the user was going to like anyway.

The same field plays opposite roles across tracks: **artist** is a taste input on Track A and the held-out label on Track B; **pixels** are an input on both, but only work on B; **engagement** is a label and a training input on A, and is not used on B.

3. Corpus construction

Frozen before any modelling, in this order, because ArtEmis coverage is the binding constraint:

1. **Artworks with ≥ 5 ArtEmis annotations** (80,030) — a 9-dimensional emotion distribution from one or two annotators is noise.
2. **Eight committed styles** (36,584), chosen for emotional balance and historical span, excluding Impressionism and Realism which would each dominate on their own.
3. **Deduplication** by perceptual hash, then exact title within artist — never by embedding distance, which would delete the easiest hits of the very method evaluated in §6.
4. **Artists with 10–200 works** (20,833 paintings, 472 artists, median 29). Below 10, recall swings wildly; above 200, a few prolific artists supply most positives for every query.
5. **One seeded shuffle, then freeze.** `item_id` indexes the corpus everywhere afterwards; the shuffle is a correctness requirement, because row order must carry no artist or style signal when ties are resolved.

4. Task, protocols and metrics

Both tracks are **top-K ranking, $K = 10$** — not rating prediction, so no RMSE appears anywhere. Every method scores every candidate; there is no pre-filtering or candidate-generation stage, so comparisons are like-for-like on the same split, candidates and metrics.

Three protocols:

- **P1 — full-catalogue ranking against the logged test set (Track A).** Rank all 20,833 items, drop what the user was shown in train/val, score the top 10 against their held-out engagements. The log is MNAR by construction, so this protocol partly rewards predicting *what the system showed people*.
- **P2 — re-ranking the uniformly-exposed holdout (Track A).** Re-rank 20 items drawn under uniform exposure, disjoint from the log. Popularity gets no free ride here.

- **P3 — item-to-item retrieval (Track B).** 2,000 sampled queries, each scored against all 20,832 other paintings, relevance is same-artist (9–197 positives per query, median 67).

P1 and P2 use different candidate sets and are **not comparable to each other**; the *gap between them* is the measurement, and it is why the simulator exists — biased-train / unbiased-test cannot be built from not-owned logs.

NDCG@10 is the headline on every rung, and every rung has a second metric that explains why it moved. NDCG says a rung gained or lost; on its own it never says what the rung did. Relevance is binary throughout, so the five accuracy metrics are the binary-relevance forms:

metric	the question it answers	its blind spot
Hit Rate@10	did <i>anything</i> relevant appear?	saturates — cannot tell one hit from ten, or rank 1 from rank 10
Precision@10	what fraction of the slots were worth showing?	ignores order entirely
Recall@10	how much of what they'd have liked did we find?	structurally capped at 10/ P
MRR@10	how far down before something useful?	sees only the first hit
NDCG@10	how many, and how high, at once?	a single summary — two very different lists can score the same

Beyond accuracy: catalogue coverage@10 and novelty in bits (across queries), average popularity (Track A), same-style share (Track B, within list), and intra-list distance plus distinct artists per list (within list). Three things that are *not* part of any metric change what it means — the candidate set (NDCG@10 over 20,833 candidates and over 20 are different quantities with the same name), the relevance definition, and K.

Every rung carries a **95% bootstrap confidence interval** on NDCG@10 and a **paired Wilcoxon test against the rung below**, on P2 for Track A and P3 for Track B. The reading rule, fixed before those numbers existed: *a rung whose interval overlaps the rung below's is reported as a rung but not claimed as a gain*, whatever the point estimates say. This matters because a paired test at $n = 11,695$ will call almost anything significant.

Four cases in this run where a single metric would have given the wrong conclusion:

- **Hit Rate saturates on P2.** Random scores 0.9303 — with 20 candidates and $K = 10$, almost every user has some positive in a random top-10.
- **Coverage without accuracy is worthless.** Normalised item-kNN reaches 85.6% coverage with NDCG@10 of 0.0015, *below Random*. It spreads widely because its scores are close to meaningless.
- **Accuracy without coverage is a product failure.** Popularity wins P1 outright while reaching 54 paintings of 20,833.
- **Novelty can be a symptom, not a virtue.** The raw engagement rate posts the highest novelty in the notebook (18.8 bits) because it ranks items with 3.4 impressions on average. It is not exploring; it is surfacing noise, and its NDCG of 0.0001 says so.

5. Methods

There is no deep learning and no training of encoders anywhere: every method is a similarity function over a fixed representation, a count, or a conjugate Bayesian update — with two exceptions, both fitted on the validation split (implicit ALS and the per-user ridge). CLIP¹ is used at inference only, a deep *representation* dropped into the existing pipeline rather than a deep *model* fitted to the data.

Random draws K items uniformly from the candidate set and fixes the scale. On Track B a stronger floor, **B1**, draws uniformly *within the query's style* — the bar emotion has to beat for “emotion adds signal” to mean anything.

Popularity scores every item by its training impression count, identically for every user. It matters because it exploits the log's MNAR structure directly: exposure was generated proportional to popularity.

Content-based turns each painting into an L2-normalised feature vector and ranks by cosine. Two vectors, because artist is a feature on one track and the answer key on the other: w_{user} (**489 dims** — 9 emotion proportions, 8 IDF-weighted style one-hots, 472 IDF-weighted artist one-hots) on Track A, and w_{item} (**17 dims**, artist excluded) on Track B. The Track A user profile is the mean of the vectors the user engaged with — the classic “average of what you liked” — whose known weakness, overspecialisation, is what coverage measures.

The per-user ridge (rung 2b) fits per-user coefficients on w_{user} using the user's *impressions*, with engaged 0/1 as the target, so it learns from what was shown and ignored as well as from what was engaged. Same signal as the content profile; the whole gap is the estimator and the negatives. λ is chosen on validation, solved in the dual for users with fewer impressions than columns.

Bayesian quality ranking asks, of the people who saw an item, how many engaged. With t_i impressions and s_i engagements, a Beta(1,1) prior gives $\theta_i \mid \text{data} \sim \text{Beta}(1 + s_i, 1 + t_i - s_i)$, and three rankers are compared: the raw rate s_i/t_i , the posterior mean $(1 + s_i)/(2 + t_i)$, and the lower 5% credible bound. The number of items tied at rank 10 is printed beside each, because that is what makes the raw rate's failure visible rather than merely arguable.

Item-kNN builds the binary item $\times$ user matrix of training engagements and scores pairs with a shrunk cosine — cosine on the binary vectors times $|U_i \cap U_j|/(|U_i \cap U_j| + \lambda)$ with $\lambda = 5$. Pairs with fewer than 2 co-engagements are zeroed and only the top 50 neighbours per item are kept. On Track A an item is scored by summing similarities to the items that user engaged with; items with no neighbourhood evidence are backfilled with the posterior mean and always ranked below every evidence-backed item, so the backfill can never manufacture a hit.

Implicit ALS sees exactly what item-kNN sees and factorises the whole matrix — “not engaged” as weak evidence of zero preference, each engagement at confidence $1 + \alpha$ — into f -dimensional user and item factors. (f, λ, α) come from a fixed 45-point grid on validation. It is a different estimator of the same signal, not a new one.

The hybrid (rung 7) z-scores the ridge and ALS scores per user over the whole catalogue and blends them, $\alpha \cdot z_{\text{ridge}} + (1 - \alpha) \cdot z_{\text{ALS}}$, with α from a 0.1 grid on validation.

¹ Contrastive-Language Image Pretraining, see also <https://openai.com/index/clip/> and <https://github.com/openai/CLIP>.

CLIP ViT-B/32, frozen, supplies cosine similarity over L2-normalised image embeddings: as the Track B method I1 (CLIP), as a Track A user profile (the same construction as the metadata profile, so the only difference is the representation), and in the Track B hybrid H1, a convex blend of the metadata and CLIP cosines. Both terms are cosines on unit vectors, so they share a scale and need no calibration.

Thompson sampling is the only *online* experiment. 500 arms; each round a simulated visitor arrives, the policy picks an arm, and the environment returns a Bernoulli engagement. Thompson draws $\tilde{\theta}_a \sim \text{Beta}(1 + s_a, 1 + t_a - s_a)$ and plays the argmax, so an arm with little data is played because it *might* be good. The comparison is against greedy, which plays the argmax of the posterior mean and stops exploring, each under two priors.

Tuning discipline. Only three settings are chosen from data — ALS’s (f, λ, α) , the ridge’s λ and the two blend weights; everything else is fixed in advance. All selection happens on validation under P1’s shape, which was committed before the results existed. A second uniformly-exposed holdout re-scores every grid point under P2’s shape to price that decision: the alternative choices are worth 0.007–0.009 on the single models and nothing on the blend, and no rung’s ordering changes. Ties are resolved by a seeded uniform draw rather than by row index, and the corpus is shuffled at freeze time, so no result can depend on the order of a file.

6. Track A — “For you”

Each rung adds exactly one new signal to the rung below, so every gap is attributable to a specific piece of information rather than to “a better model”. NDCG@10 is on **P2**; Δ is against the rung below.

#	rung	signal added	NDCG@10	95% CI	Δ	Wilcoxon p
0	Random	—	0.4015	0.398–0.405	—	—
—	Content (CLIP pixels)	512-d image embedding	0.4159	0.412–0.420	+0.0144	1e-8
1	Popularity (baseline)	impression count	0.4552	0.451–0.459	+0.0537	4e-106
2	Content (metadata)	emotion, style, artist	0.5187	0.515–0.523	+0.0635	2e-139
2b	Content, per-user ridge	rung 2’s metadata, fitted on impressions	0.5972	0.594–0.601	+0.0786	<1e-300
3	Quality, raw rate	engagements / impressions	0.5244	0.520–0.529	+0.0057	0.015 — <i>CIs overlap</i>
4	Quality, posterior mean	+ a Beta(1,1) prior	0.5317	0.528–0.536	+0.0073	4e-41 — <i>CIs overlap</i>
5	Quality, lower 5% bound	+ a credible interval	0.5385	0.534–0.543	+0.0068	6e-17 — <i>CIs overlap</i>
6	Item-kNN	co-engagement	0.5717	0.568–0.576	+0.0332	1e-63
6b	Implicit ALS	rung 6’s signal, latent factors	0.5984	0.594–0.603	+0.0267	8e-48
7	Hybrid (ridge + ALS) (improvement)	2b and 6b blended	0.6324	0.629–0.637	+0.0339	2e-179
—	Oracle	true utility	0.8919	—	—	—

The ladder closes **47.1%** of the 0.4904 gap between the random floor and the oracle ceiling: real progress, and most of the distance still unclosed. Seven of the ten steps are claimed; the three quality rungs are not.

Rungs 3–5 are one rung split three ways. The signal is identical — engagement rate — and only the estimator changes, so +0.014 from raw rate to lower bound is a pure measurement of what Bayesian shrinkage is worth. No single step is claimed, but the end-to-end gap is: raw rate’s interval and the lower bound’s do not touch. The legible effect is not in NDCG at all — it is the tie count at rank 10, which drops from **331 to 1** the moment the prior is added. The raw rate fills its top with items engaged 2-of-2 that nobody has really seen, and the tie-break, not the estimator, picks the list.

Rung 6 is where behaviour earns its place. The simulator’s utility contains a 16-dimensional latent term unrelated to any observable feature; no content model can reach it, and co-engagement is the only trace it leaves. Item-kNN beats the content profile by +0.0530 on P2 and by an order of magnitude on P1. Implicit ALS, estimating the same signal, then beats item-kNN on both protocols — with a tuned factor count of **16**, exactly the dimension of the hidden term it is recovering, and anything wider fits noise. Its cost is concentration: 671 distinct paintings across 10,448 top-10 lists, at an average recommended popularity of 1,213 impressions against item-kNN’s 995.

Rung 2b reorders the reading of the whole ladder. The per-user ridge uses rung 2’s metadata and nothing else, but is fitted per user on *impressions* — so it learns from what the user was shown and ignored, not only from what they engaged with. It reaches 0.5972, the largest single step on the ladder, and lands inside implicit ALS’s interval. A direct paired test makes it a tie in the strict sense: $\Delta = +0.0012$ for ALS, **p = 0.58**. And it ties at a completely different product cost — **5,925 distinct paintings** against ALS’s 671, at an average popularity of 74, the catalogue’s own average, against 1,213. Against the same metadata with an estimator of equal strength, behaviour’s accuracy advantage on the unbiased protocol is gone and its concentration cost remains.

Rung 7 resolves the tie. The two estimators fail differently — the ridge cannot see the hidden factor, ALS cannot see a user with no engagements — so a convex blend of their per-user z-scores at $\alpha = 0.3$, chosen on validation, reaches **0.6324**: +0.0339 over ALS and +0.0351 over the ridge, both claimed, and the largest single gain on the ladder above rung 2. It is the improvement this project delivers on Track A, and it is the cleanest evidence that the hidden factor *is* there and *is* reachable — just not by either estimator alone. What it does not inherit is the ridge’s coverage: 834 distinct paintings. So the product question survives the accuracy question — the hybrid for accuracy, the ridge for a catalogue that actually gets shown.

CLIP is not a rung — it is a rung that failed to climb. Used as a user profile it sits second from the bottom, 0.10 below the metadata content model, and on P1 it scores 0.0007 against Random’s 0.0008, i.e. *below chance* (a later paired test puts it at $p = 0.72$ against Random: it is not merely below chance, it *is* chance there). This was predicted in writing before the run, for a stated reason: the simulated utility is assembled from emotion, style, artist and a hidden factor, and **not one of those is a pixel statistic**. Sweeping the metadata weight from 0 to 1 raises NDCG monotonically, so the optimum is pure metadata and pixels contribute nothing at any mixing weight.

The ladder is not monotone on the other protocol. On **P1** the same rungs give Popularity 0.0488, Content 0.0092 and the quality rankers 0.0033 — content and quality fall *below* the baseline they beat on P2. Only the behavioural rungs win both.

7. Track B — “More like this”

#	rung	NDCG@10	95% CI	Δ	p	same-style	intra-list dist.
0	B0 Random	0.0042	0.003–	—	—	0.150	0.392

#	rung	NDCG@10	95% CI	Δ	p	same-style	intra-list dist.
			0.005				
1	B1 Random within style (<i>baseline</i>)	0.0262	0.023–0.029	+0.0220	2e-69	1.000	0.347
2	B2 Content: emotion + style	0.0431	0.040–0.047	+0.0169	2e-13	1.000	0.333
3	B3 Item-kNN	0.0100	0.008–0.012	–0.0331	1e-57	0.355	0.377
3b	B4 Implicit ALS	0.0049	0.004–0.006	–0.0050	1e-7	0.171	0.385
4	I1 CLIP ViT-B/32	0.3306	0.318–0.345	+0.2875	1e-227	0.630	0.172
5	H1 Hybrid (metadata + CLIP) (<i>improvement</i>)	0.4000	0.387–0.415	+0.0694	1e-88	1.000	0.180

Every step here is claimed: no two adjacent intervals touch. Track B has only 2,000 units against Track A's 11,695, so its intervals are wider — and its effects are an order of magnitude larger, which is why they survive anyway.

B1 → B2 is the finding the project was designed around. Both return same-style items 100% of the time, so the entire $1.65\times$ gain comes from emotion proportions re-ordering paintings *within* a style. It is also the smallest claimed gain on the track, so it is real, and it is modest. A split-half reliability check bounds it further: the emotion vectors are noisy measurements (split-half correlations of 0.07 to 0.38; two half-vectors of the same painting agree on the dominant emotion only 35% of the time), and two half-annotator B2 lists overlap by **0.8%**. Yet **69% of the B1 → B2 gain survives on half the evidence**. What B2 exploits is the coarse tendency for an artist's works to share a mood, which two or three annotators are enough to catch; what it cannot deliver is a *specific* list a second crowd would reproduce. **The metric is robust, the lists are not.**

Rung 3 goes down, not up. Co-engagement is the best signal on Track A and among the worst here: a visitor's taste spans artists, and B3's same-style share of 0.355 shows it is not even recovering the style. B4 goes further down, landing within a whisker of Random with a same-style share of 0.171 against the random base rate of 0.150 — sixteen factors fitted to *taste* contain no authorship and almost no style. The ladder metaphor holds only within a task.

CLIP does not edge past metadata — it laps it. NDCG@10 0.3306 against B2's 0.0431 is **7.7×**; Precision@10 goes 0.0433 → 0.2984 and MRR 0.1022 → 0.5546, so the first same-artist hit typically lands around rank 2 instead of rank 10. Artist is hidden from every method, the query set and tie-breaking are unchanged, and Track B contains no synthetic data. Note that CLIP's same-style share is 0.630, not 1.000: it finds the artist directly rather than by first narrowing to the style, which is a different retrieval behaviour from every metadata baseline.

The hybrid beats pure CLIP, which is the textbook argument for hybridisation arriving with a measured margin. Metadata knows the crowd's emotional reading and the style taxonomy, CLIP knows composition, palette and texture, and the two fail differently. Re-selecting the blend weight on a tuning half of the queries and scoring it on the other half returns the same $\alpha = 0.25$ and a small further gain whose intervals overlap — so it is reported, not claimed. The more useful result is the shape of the curve: NDCG is flat between $\alpha = 0.05$ and 0.45, so almost any metadata weight short of dominance does the same job, and the only cliff is at $\alpha \rightarrow 1$ where CLIP is switched off.

Read that contrast against Track A, where the same embeddings are worthless. A representation can be transformative against real labels and useless against simulated ones — the strongest evidence in the project for keeping the two tracks apart. A study reporting only Track A would have concluded that CLIP does not work for art recommendation.

The cost is real and was nearly missed. Series collapse was predicted in advance, with coverage@10 and novelty named as the diagnostics. Both look *healthy* for CLIP (0.515 coverage, 15.6 bits) because they are cross-query metrics and the failure is within-list. Intra-list distance halves, $0.333 \rightarrow 0.172$, on 99.85% of queries. The prediction was right and the instrument was wrong, which is the more useful lesson. The hybrid recovers same-style discipline (1.000) and a little diversity (+0.008).

8. Research questions, answered

RQ1 — Does the evaluation protocol decide the winner? Yes. Popularity beats the content model on the exposure-biased log (0.0488 against 0.0092 on P1) and loses to it on the uniformly-exposed holdout (0.4552 against 0.5187). Its average recommended popularity of 2,070, against the catalogue's 72, shows the mechanism, and its coverage of 54 items shows the product cost. On P1 it is partly predicting the logging policy rather than taste.

RQ2 — Engagement quality, or exposure volume? Protocol-dependent, which is RQ1 arriving again. On P2 all three quality rankers beat Popularity with room to spare; on P1 the raw rate is the worst method in the notebook (0.0001, below Random) for exactly the predicted reason — 331 items tie at rank 10 and the tie-break draws the list. Shrinkage is worth ~ 0.014 end to end, and the data cannot say which of the two shrunk estimators is better.

RQ3 — Does behaviour beat metadata? Split. Against a positives-only content profile, yes, twice over: item-kNN +0.0530 on P2 and an order of magnitude on P1, then implicit ALS above item-kNN on both. But against the *same metadata fitted per user on impressions* (rung 2b), the answer on the unbiased protocol is **no** — 0.5972 against 0.5984, $p = 0.58$, at nine times the coverage — and on the logged protocol **yes**, by an order of magnitude. The hidden factor's contribution to what a user would click, given a fair candidate set, is smaller than the simulator's R^2 suggested. What behaviour buys decisively is P1. The two are complementary rather than redundant: the blend beats both by +0.034.

RQ4 — What actually drives “more like this”? In order: pixels, style, emotion — behaviour last, and by a wide margin. $B1 \gg B0$ (6 $\times$), $B2 > B1$ (1.65 $\times$), and both behavioural methods lose to random-within-style.

RQ5 — Do pixels beat metadata? Yes on Track B (7.7 $\times$, and 0.4000 once metadata is blended back in), no on Track A (0.4159 against metadata's 0.5187 on P2, below Random on P1). The cost is a halved intra-list distance.

RQ6 — Does exploration pay? In circulation, yes; in reward, only briefly. Thompson sampling is the only non-random policy that keeps the whole catalogue in play — 500 arms at both horizons, against greedy's 399 at 20,000 rounds collapsing to 14 at 100,000 with every play on one arm. On reward, greedy wins under Beta(1,1) at both horizons. The reason is the prior: only 23 of 500 arms have a true rate above the Beta(1,1) mean of 0.5, so Thompson's optimism keeps sampling arms almost all worse than its incumbent. Under the pessimistic Beta(1,3), Thompson wins the short horizon and greedy the long one. **The prior matters more than the policy.**

9. Limitations and what the numbers cannot support

Not supported. Any absolute claim about real visitors on Track A. Those numbers characterise this simulator, and the conclusions that depend on stipulated weights are swept to show where they reverse.

Not tested at all. Whether the recommended paintings actually evoke awe, contentment or amusement in a human being. ArtEmis supplies a crowd-annotated *proxy* for emotional response, and the simulated user reacts to that proxy by construction. Closing that loop needs a study with real participants — the honest next step, and not more modelling.

- **The user layer is simulated,** and Track A's circularity is bounded and measured rather than assumed away.
- **Curatorial proxy ceiling on Track B.** A visually apt recommendation by a different artist scores as a miss, so absolute precision is a lower bound on real quality; only relative comparisons are meaningful.
- **Series collapse.** A strong visual model returns near-identical works from one series — ten Monet haystacks, a run of Rothko colour fields. All are same-artist, so Precision@10 rises while the module gets worse for exploratory browsing. Perceptual hashing removes literal re-scans, not series.
- **Aggregation and annotator-pool bias.** Emotion distributions pool ~5.8 annotators, so they measure population-average *response*, not individual preference. ArtEmis annotators are non-expert English-speaking crowdworkers, a culturally narrow sample.
- **Limited external validity.** Neither dataset contains interaction data. The corpus is Western-canon-skewed, and the 10–200 artist band removes Van Gogh, Picasso, Dürer, Warhol and most other canonical names.
- **A reproducibility failure worth recording.** The CLIP checkpoint tag originally pinned builds a non-QuickGELU model and loads QuickGELU-trained weights into it, emitting a warning and proceeding. The two tags give embeddings at cosine 0.957 of each other — close enough to look correct — while their nearest neighbours agree only 60.9% of the time. For a retrieval evaluation that is a different experiment, not a rounding difference.

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Learning experience

This assignment provided an opportunity to gain a deeper understanding of the conceptual and practical implementation of a recommender system. The process of selecting an appropriate dataset and deciding which recommendation approach to implement was undertaken only after acquiring a general understanding of the concepts introduced during the lectures.

The implementation phase required substantial support from AI tools, as part of the steps were not covered in the course notebooks or classroom exercises. This aspect had a significant impact on the project, particularly because it was developed individually and demanded a high degree of autonomous problem-solving.

The project was developed in three main phases:

1. Planning: reviewing relevant literature, identifying suitable datasets, defining the project workflow, and selecting evaluation metrics.
2. Implementation: developing and testing the code, followed by analyzing and discussing the results.
3. Assembly: answering the research questions, discussing the project's limitations, and revising the report for clarity and consistency.

The final report serves both as a means of communicating the project's objectives, methodology, and findings to the reader and as a tool for consolidating the author's learning process. Throughout the project, periods of progress were accompanied by moments of uncertainty, which ultimately contributed to a deeper understanding of the subject matter.

One challenge was the gap between the project requirements and the examples provided during the course. Because the project differed substantially from the classroom exercises, it was not always possible to rely solely on the lecture materials. Consequently, AI-assisted guidance became necessary to bridge missing implementation details and to validate the overall direction of the work. A project example more closely aligned with the assignment requirements would likely have facilitated a smoother learning process and strengthened the connection between theoretical concepts and practical application. However, the author drafted the assignment according to a personal curiosity through art datasets, to be implemented in the GLAM (Galleries, Libraries, Archives, Museums) sector.

Overall, despite these challenges, the project represented a positive learning experience in an autonomous working environment. It fostered both technical skills and independent problem-solving abilities while providing valuable insight into the design, implementation, and evaluation of recommender systems.